

Supersonic Flow around Airfoils [8.5, Chapter 10]

Objective:

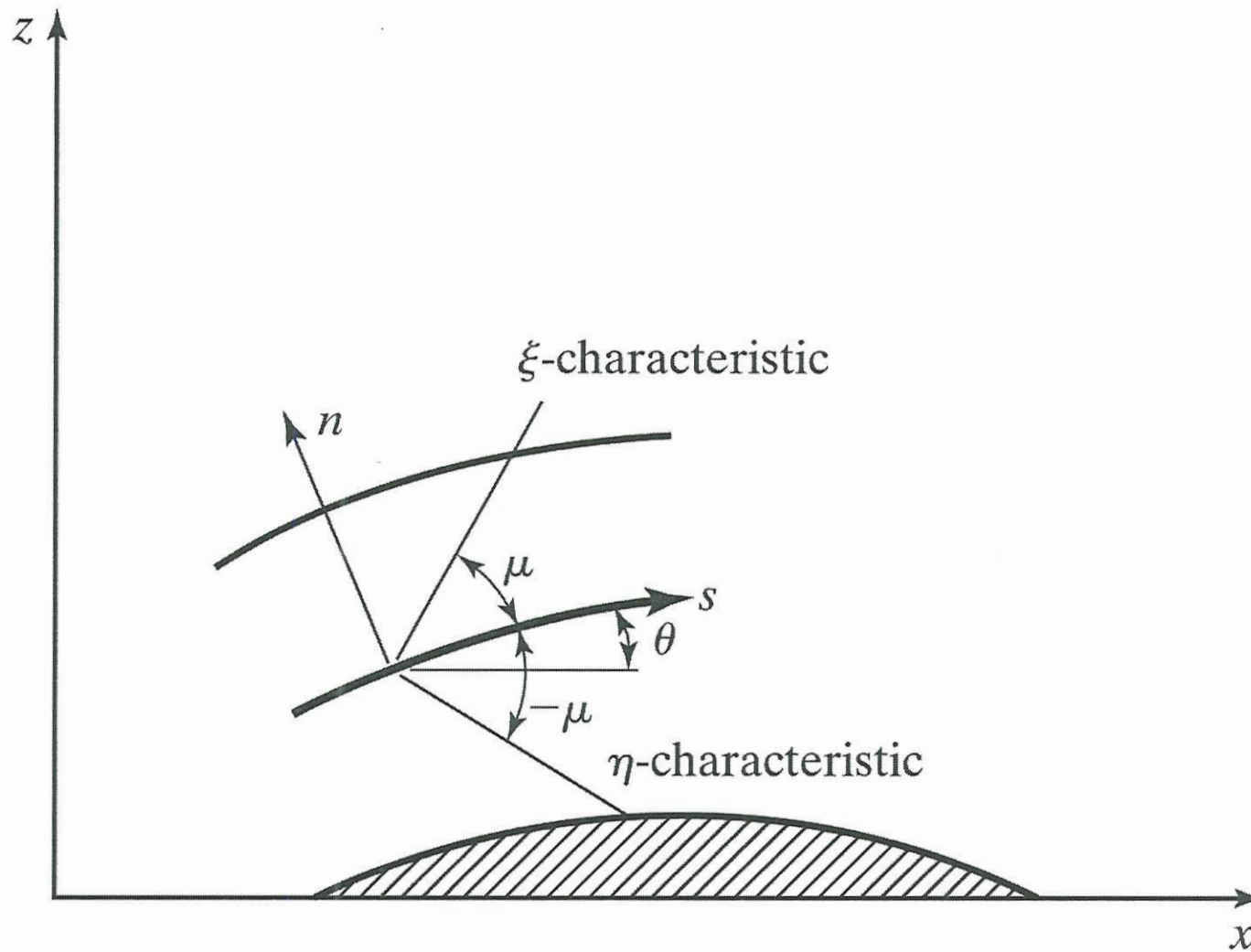
Analyze (isentropic) expansion

Analyze supersonic airfoils using shock/expansion wave patterns

Analyze supersonic airfoils using linearized theory

Compare the above two methods (including limitations and assumptions)

Characteristic Eqs & Prandtl-Meyer Expansion Waves [8.5]



Conservation laws in polar coordinates

continuity
$$\frac{\partial(\rho U)}{\partial s} + \rho U \frac{\partial \theta}{\partial n} = 0 \quad (8.51)$$

s-momentum
$$\rho U \frac{\partial U}{\partial s} + \frac{\partial p}{\partial s} = 0 \quad (8.52)$$

n-momentum
$$\rho U^2 \frac{\partial \theta}{\partial n} + \frac{\partial p}{\partial n} = 0 \quad (8.53)$$

Assume isentropic flow

From (8.26)
$$\frac{\partial p}{\partial \rho} = a^2 \quad (8.54)$$

Putting in (8.51)
$$\frac{\partial p}{\partial s} \frac{M^2 - 1}{\rho U^2} + \frac{\partial \theta}{\partial n} = 0 \quad (8.56a)$$

- continuity equation

From (8.53) and (8.55)
$$\frac{\partial p}{\partial \xi} + \frac{\rho U^2}{\sqrt{M^2 - 1}} \frac{\partial \theta}{\partial \xi} = 0$$

directional derivative

Along
$$\frac{\partial n}{\partial s} = \tan \mu = \frac{1}{\sqrt{M^2 - 1}} \quad \xi - \text{characteristics - left running}$$

$$\mu = \sin^{-1} \left(\frac{1}{M} \right)$$

Also
$$\frac{\partial p}{\partial \eta} - \frac{\rho U^2}{\sqrt{M^2 - 1}} \frac{\partial \theta}{\partial \eta} = 0 \quad (8.56b) \quad ; \quad \frac{\partial n}{\partial s} = -\tan \mu$$

\eta - characteristics - right running

8.56a,b: relations between pressure and flow direction

Euler's equation: $dp = -\rho U dU$

From (8.56a)
$$\frac{dU}{U} = \frac{d\theta}{\sqrt{M^2 - 1}} \quad (8.57)$$

adiabatic flow
$$\left(\frac{U}{a_t}\right)^2 = M^2 \left(1 + \frac{\gamma - 1}{2} M^2\right)^{-1} \quad (8.58)$$

Differentiating and substituting to (8.57)

$$d\theta = \frac{\sqrt{M^2 - 1} dM^2}{2M^2 \{1 + [(\gamma - 1)/2] M^2\}} \quad (8.59)$$

Integrating: $\theta = \nu + \text{constant}$; ν is the Prandtl-Meyer function

$$\nu = \sqrt{\frac{\gamma + 1}{\gamma - 1}} \tan^{-1} \sqrt{\frac{\gamma - 1}{\gamma + 1} (M^2 - 1)} - \tan^{-1} \sqrt{M^2 - 1} \quad (8.60)$$

Note: $\nu(M) = 0$ when $M = 1$

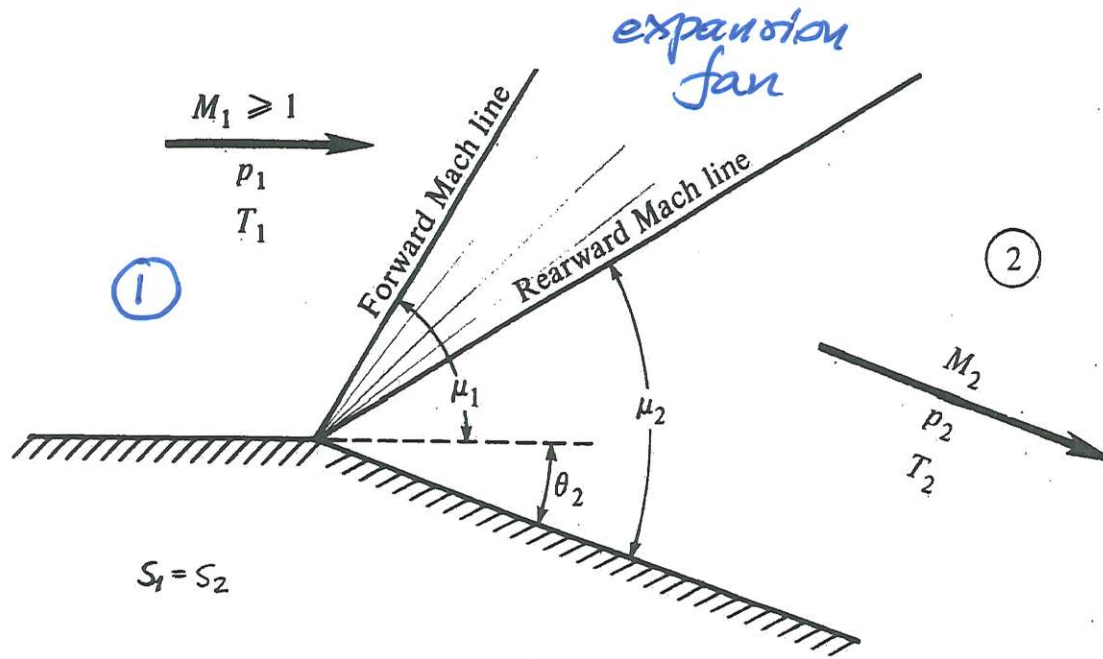
ν is tabulated as a function of M (Table 8.3)

Along left running characteristics: $v - \theta = R$

Along right running characteristics: $v - \theta = Q$

Example – 8.3 Supersonic thin airfoil [see text] *- not important*

For a finite expansion (for left running characteristics; top surface),
isentropic flow $\Delta v = \Delta \theta$



Process:

find $v(M_1)$ from Table 8.3

find $v(M_2)$ from

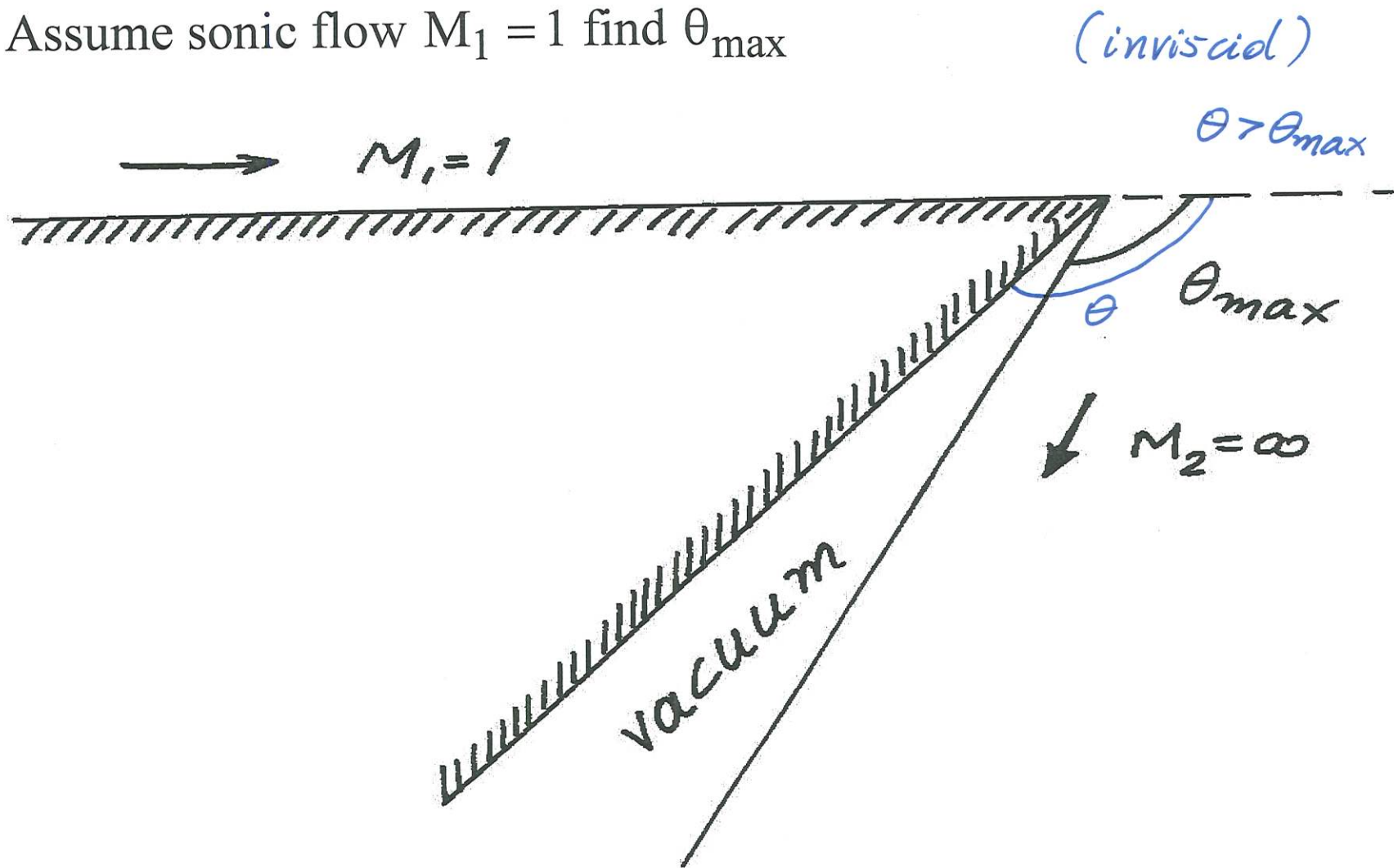
$$\theta = v(M_2) - v(M_1)$$

find M_2 from Table 3

find p_2, T_2 for isentropic
flow (Table 8.1)

$$M_2 > M_1 \quad ; \quad P_2 < P_1$$

Assume sonic flow $M_1 = 1$ find $\theta_{\max}$



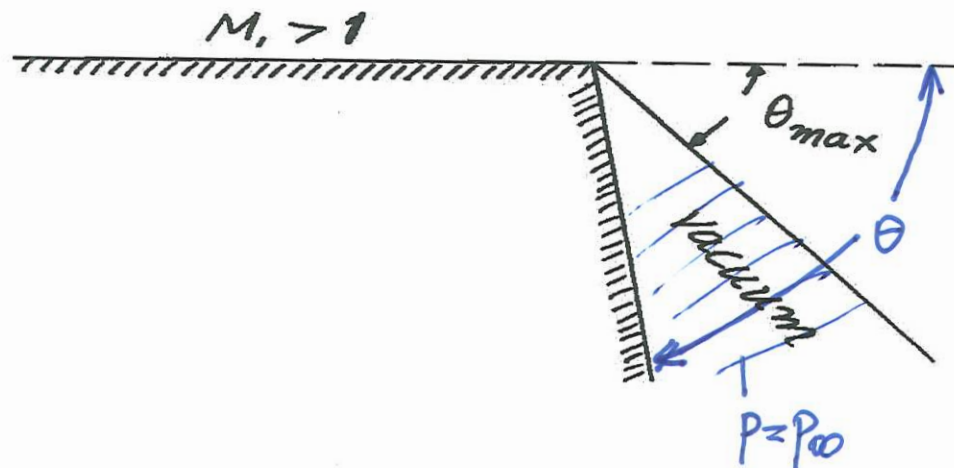
$$\lim_{M_2 \rightarrow \infty} v_2 = \lim \left(\sqrt{\frac{\gamma+1}{\gamma-1}} \tan^{-1} \sqrt{\frac{\gamma-1}{\gamma+1}} M_2^2 - 1 - \tan^{-1} \sqrt{M_2^2 - 1} \right)$$

$$= \sqrt{\frac{\gamma+1}{\gamma-1}} \frac{\pi}{2} - \frac{\pi}{2} = \left(\sqrt{\frac{\gamma+1}{\gamma-1}} - 1 \right) \frac{\pi}{2} = 2.277 \text{ rad} = 130^\circ.45$$

$$\theta_{\max} = v(M_2) - v(M_1)$$

$\downarrow$ $\downarrow$
 $130^\circ.45$ 0 for $M_1 = 1$

for any flow a $\theta_{\max}$ can be found e.g.

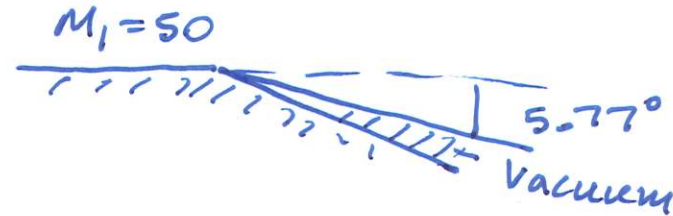


tables

$$M_1 = 5 \rightarrow v(M_1) = 76^\circ.92 \Rightarrow \theta_{\max} = 53^\circ.53$$

$$M_1 = 50 \Rightarrow v(M_1) = 124^\circ.7 \Rightarrow \theta_{\max} = 5^\circ.77$$

as



$$M_1 \uparrow \Rightarrow \theta_{\max}$$

$$M_1 \rightarrow \infty \Rightarrow \theta_{\max} \rightarrow 0$$

no expansion

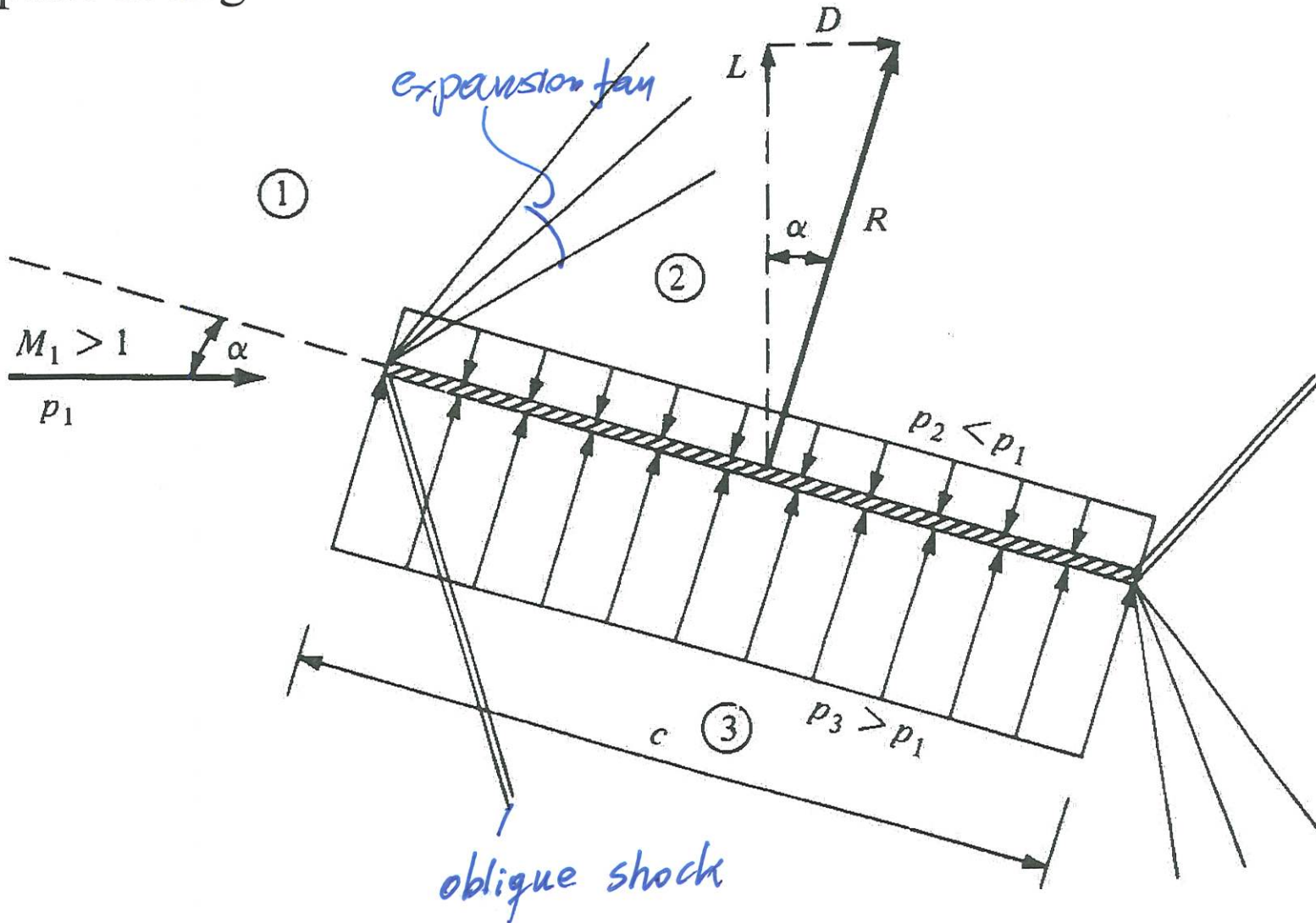


(flow cannot accelerate anymore)

*- see Newtonian theory for hypersonic flow
ch. 12*

Oblique shocks + expansion waves [10.1]

flat plate at angle α



Total force/unit depth

$$R' = (p_3 - p_2) c$$

$$\Rightarrow L' = (p_3 - p_2) c \cos \alpha$$

$$D' = (p_3 - p_2) c \sin \alpha \neq 0$$

p_2 and p_3 can be found

shock wave drag $D' \neq 0$

$$\Rightarrow \frac{L}{D} = \frac{C_L}{C_D} = \frac{\cos \alpha}{\sin \alpha}$$

Example: $M_1 = 2.6$ evaluate C_ℓ , C_d for 5° , 15° , 30°

for $M_1 = 2.6$, $\delta = 5^\circ$ from δ - θ - M diagram $\theta = 26.5^\circ$

$$M_{n_1} = M_1 \sin \theta = 2.6 \times \sin 26.5^\circ =$$

from normal shock tables: for $M_{n_1} = 1.16$ $\frac{p_3}{p_1} = 1.403 > 1$

from isentropic tables: $M_1 = 2.6$ $\frac{p_{t1}}{p_1} = 19.95$

from expansion tables: for $M_1 = 2.6$ $\nu_1 = 41.41^\circ$

$$\nu_2 = \nu_1 + \theta = 41.41 + 5 = 46.41^\circ \Rightarrow M_2 = 2.83 \quad (M_2 > M_1)$$

from isentropic tables: for $M_2 = 2.83$ $\frac{p_{t2}}{p_2} = 28.43$

expansion fan
($p_{t1} = p_{t2}$ isentropic)

$$\frac{p_2}{p_1} = \frac{p_2}{p_{t2}} \frac{p_{t2}}{p_{t1}} \frac{p_{t1}}{p_1} = \frac{1}{28.43} \cdot 1 \cdot (19.95) = 0.7022$$

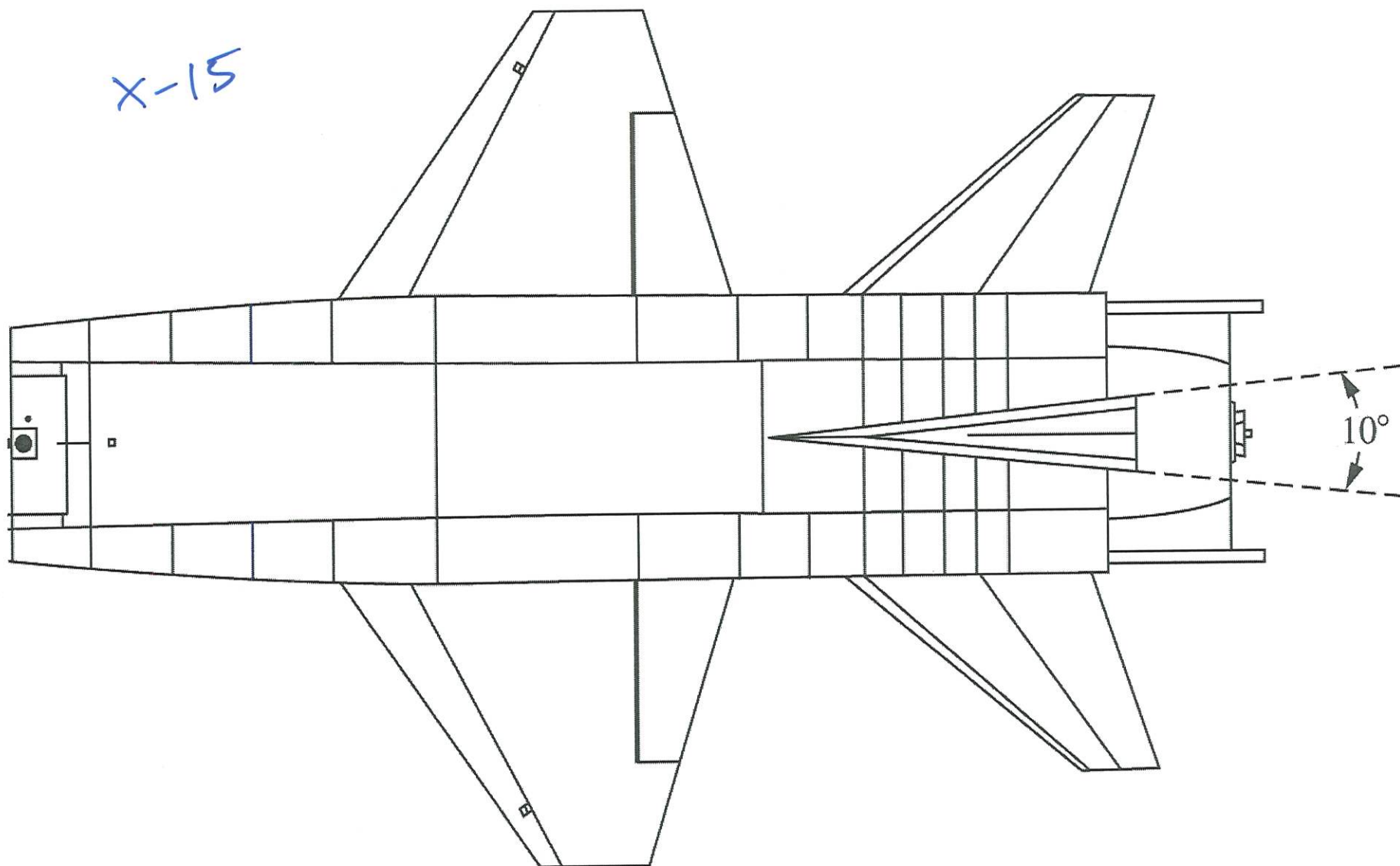
$$q_\infty = \frac{1}{2} \gamma p_\infty M_\infty^2 \quad C_l = \frac{2}{\gamma M_1^2} \left(\frac{p_3}{p_1} - \frac{p_2}{p_1} \right) \cos \alpha = 0.148$$

$$C_d = \frac{2}{\gamma M_1^2} \left(\frac{p_3}{p_1} - \frac{p_2}{p_1} \right) \sin \alpha = C_l \tan \alpha = 0.0129$$

$$\alpha = 5^\circ \rightarrow C_\ell = 0.148 \quad C_d = 0.0129$$

$$\alpha = 15^\circ \rightarrow C_\ell = 0.452 \quad C_d = 0.121$$

$$\alpha = 30^\circ \rightarrow C_\ell = 1.19 \quad C_d = 0.687$$



Example: [The x-15 and the wedge tail] Consider $M=7$ flow over a flat plate at an angle of attack of 10° and a 10° wedge. Compare the C_ℓ coefficients.

a) Flat plate; top: App C for $M=7$: $v_1 = 90.97^\circ$

$$v_2 = v_1 + \alpha = 100.97$$

From tables $M_2 = 9.56$

From isentropic tables $p_{t1}/p = 0.414 \times 10^4$; $p_{t2}/p = 0.33 \times 10^5$

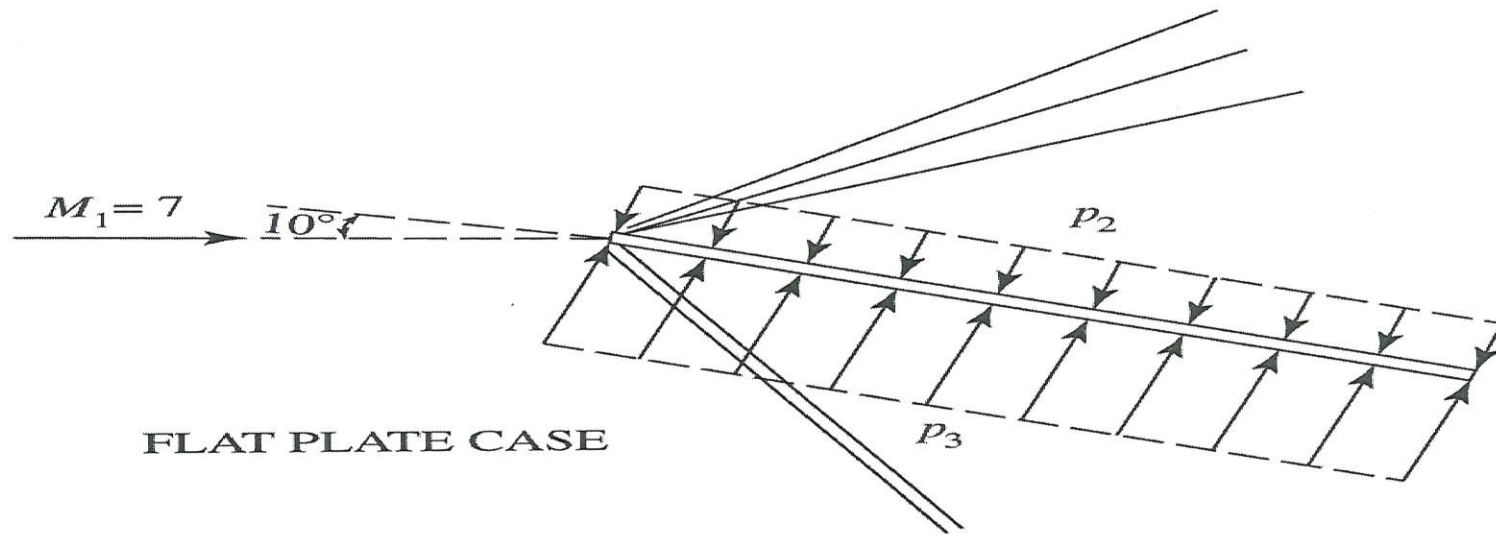
$$\text{Thus } \frac{p_2}{p_1} = \frac{p_{t1}}{p_{t2}} \frac{p_1}{p_2} = \frac{0.414 \times 10^4}{0.33 \times 10^5} = 0.1255$$

Bottom: from δ - θ - M for $M_1=7$, $\delta=10^\circ$ we get: $\theta = 16.5$

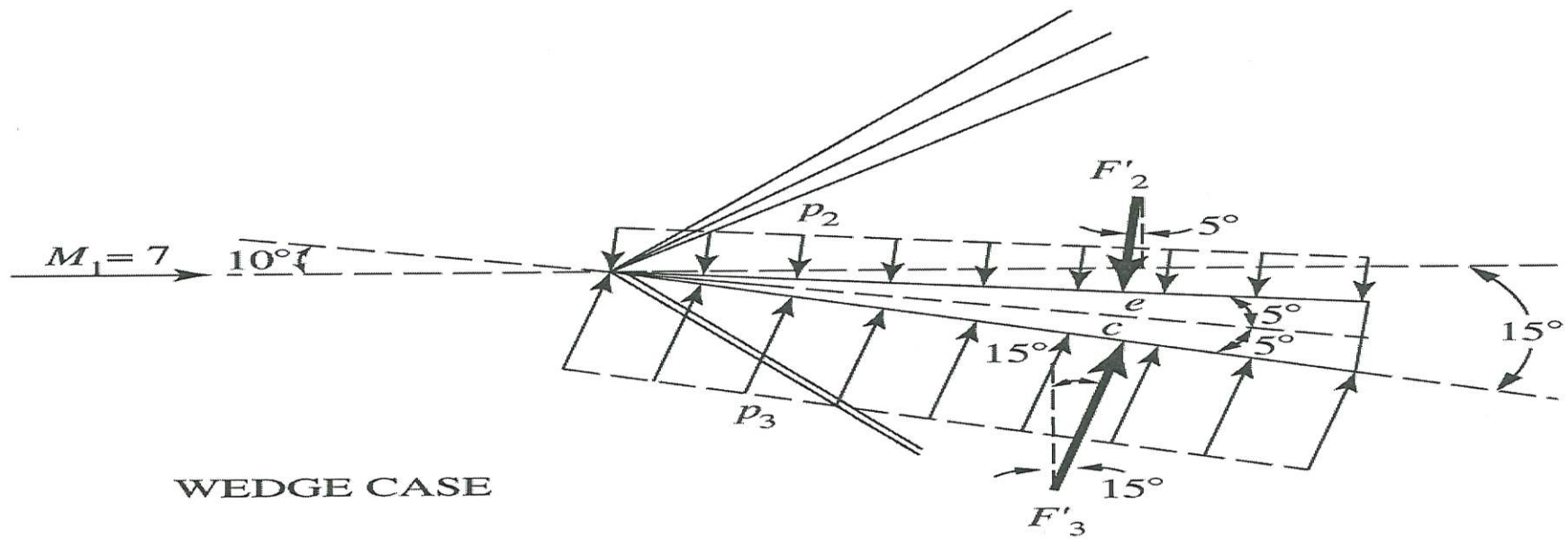
$$M_{n1} = M_1 \sin \theta = 7 \times \sin(16.5) = 1.99$$

From shock tables for $M_{n1} = 1.99$ we get $p_3/p_1 = 4.45$

$$\text{For the } C_\ell \text{ is: } \Rightarrow C_l = \frac{2}{\gamma M_1^2} \left(\frac{p_3}{p_1} - \frac{p_2}{p_1} \right) \cos \alpha = \frac{2}{1.4 \times 7^2} (4.45 - 0.1255) = 0.126$$



(a)



(b)

Note: Angles not to scale

b) Wedge; top: $v_2 = v_1 + \underset{(10-5)}{5} =$; From tables $M_2 = 8.1$ ^{expansion}

From isentropic tables $p_{t2}/p_2 = 1.0897 \times 10^4$

From geometry $\ell = c/\cos 5^\circ = 1.004c$

$$\text{Force/unit span (top)} F'_2 = p_2 \ell = \left(\frac{p_{t1}/p_1}{p_{t2}/p_2} \right) p_1 \ell = \left(\frac{0.414 \times 10^4}{1.0897 \times 10^4} \right) p_1 \ell = 0.38 p_1 \ell$$

(from before $p_{t1}/p = 0.414 \times 10^4$)

Bottom: from δ - θ - M for $M_1 = 7$, $\delta = 15^\circ$ we get: $\theta = 23.5$
^{"10+5}

$$M_{n1} = M_1 \sin \theta = 7 \times \sin(23.5) = 2.79$$

From shock tables for $M_{n1} = 2.79$ we get $p_3/p_1 = 8.915$

Thus $F'_3 = p_3 \ell = (p_3/p_1)p_1 \ell = 8.915 p_1 \ell$

Lift: $L' = F'_3 \cos 15^\circ - F'_2 \cos 5^\circ = (0.9659)(8.915)p_1 \ell$
 $- (0.9962)(0.38)p_1 \ell$
 $= 8.232 p_1 \ell = 8.265 p_1 c$
 $(\ell = 1.004c)$

Lift coefficient: $C_\ell = \frac{L'}{(\gamma/2)p_1 M_1^2 c} = \frac{8.265 p_1 c}{(\frac{1.4}{2})p_1 7^2 c} = 0.241$

Higher for wedge. Why? bottom surface has inclination 15° vs $10^\circ \leadsto$

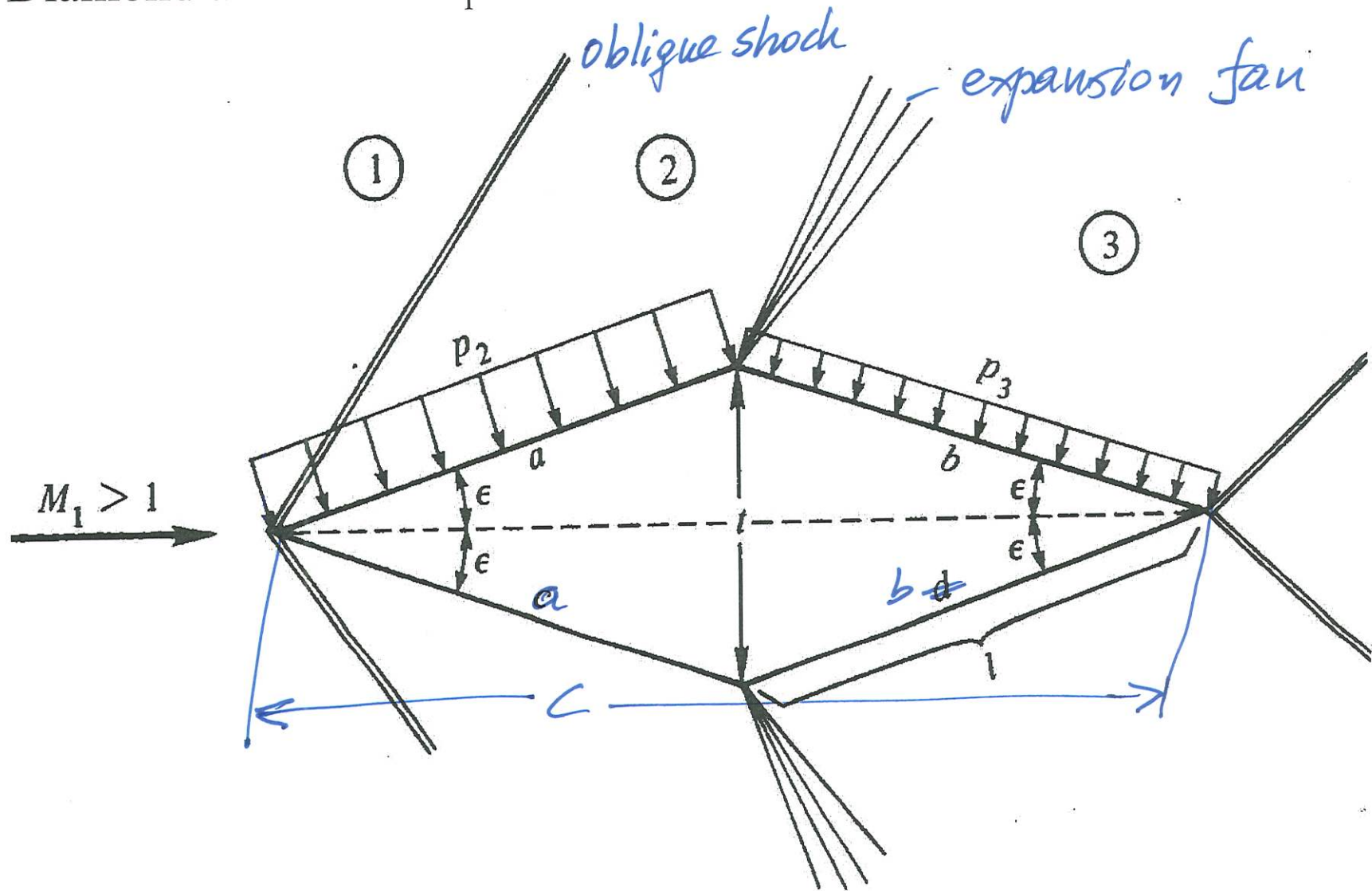
Stronger shock $\leadsto$ higher $\frac{p_3}{p_1} \leadsto$

(8.915 vs 4.45)

higher lift

(expansion on the top has small effect)

Diamond airfoil: $M_1 > 1$ $\alpha = 0$



drag/unit depth

propulsive force

$$D' = 2(p_2 \ell \sin \varepsilon - p_3 \ell \sin \varepsilon) = (p_2 - p_3) t \quad (\text{shock wave drag})$$

to get to
horizontal

$$t = 2\ell \sin \varepsilon$$

upper + lower surface

$$C_d = \frac{D'}{qc} \quad \left(\frac{l}{c} = \frac{1}{2 \cos \varepsilon} \right)$$

$$\Rightarrow C_d = \frac{2}{\gamma M_1^2} \frac{t}{c} \left(\frac{p_2}{p_1} - \frac{p_3}{p_1} \right) \neq 0$$

shock wave
drag

$$p_2 > p_1$$

oblique shock

$$p_3 < p_1$$

expansion fan

$$C_l = 0 \quad (\alpha = 0; \text{symmetry})$$

Linear Theory [10.1]

Starting with Euler's equation

$$dp = -\rho U dU \Rightarrow p - p_\infty = -\rho U (U - U_\infty)$$

Then 8.57 becomes:

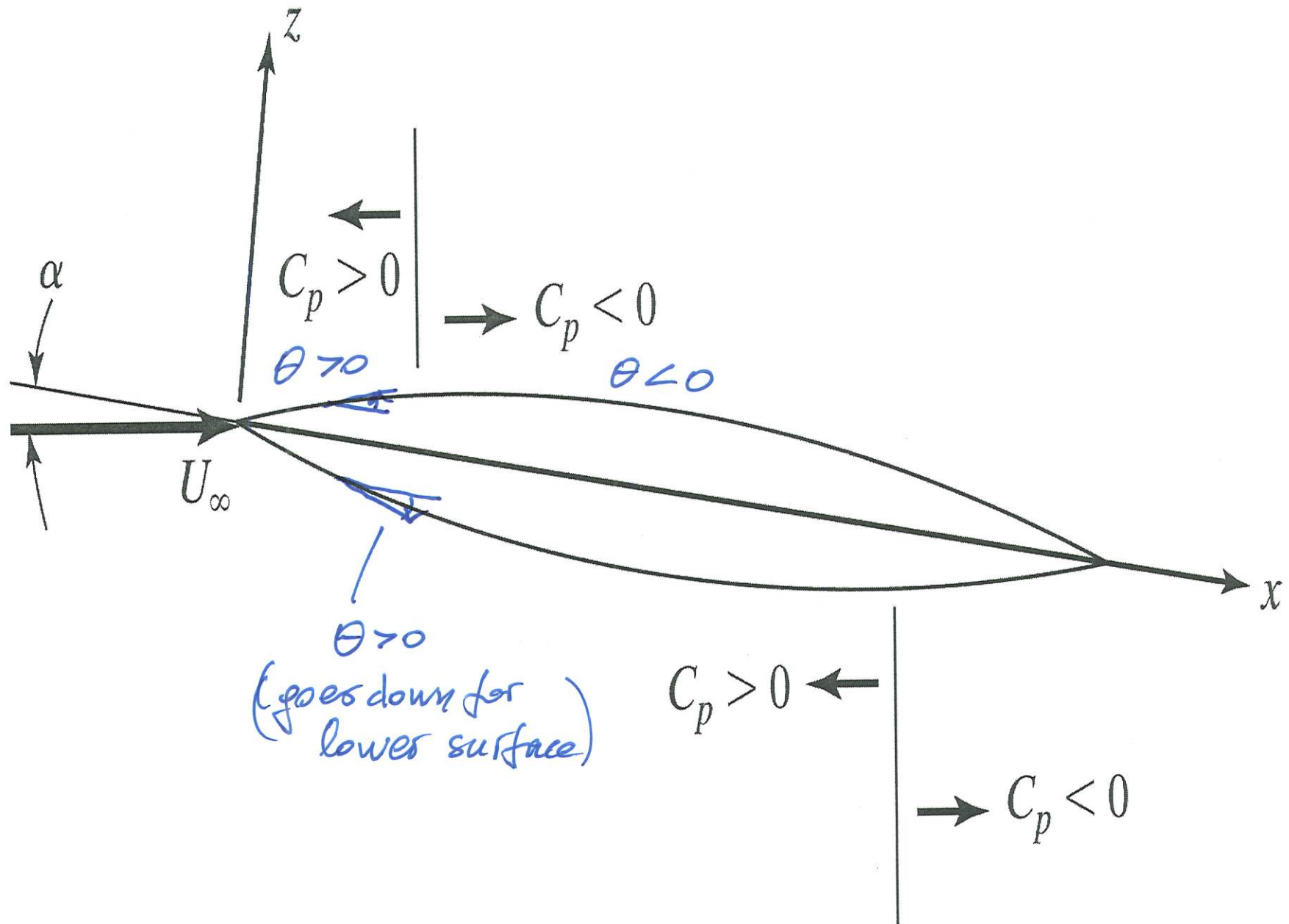
$$\frac{dU}{U} = \frac{d\theta}{\sqrt{M_\infty^2 - 1}} \Rightarrow \frac{U - U_\infty}{U} = \frac{\theta}{\sqrt{M_\infty^2 - 1}}$$

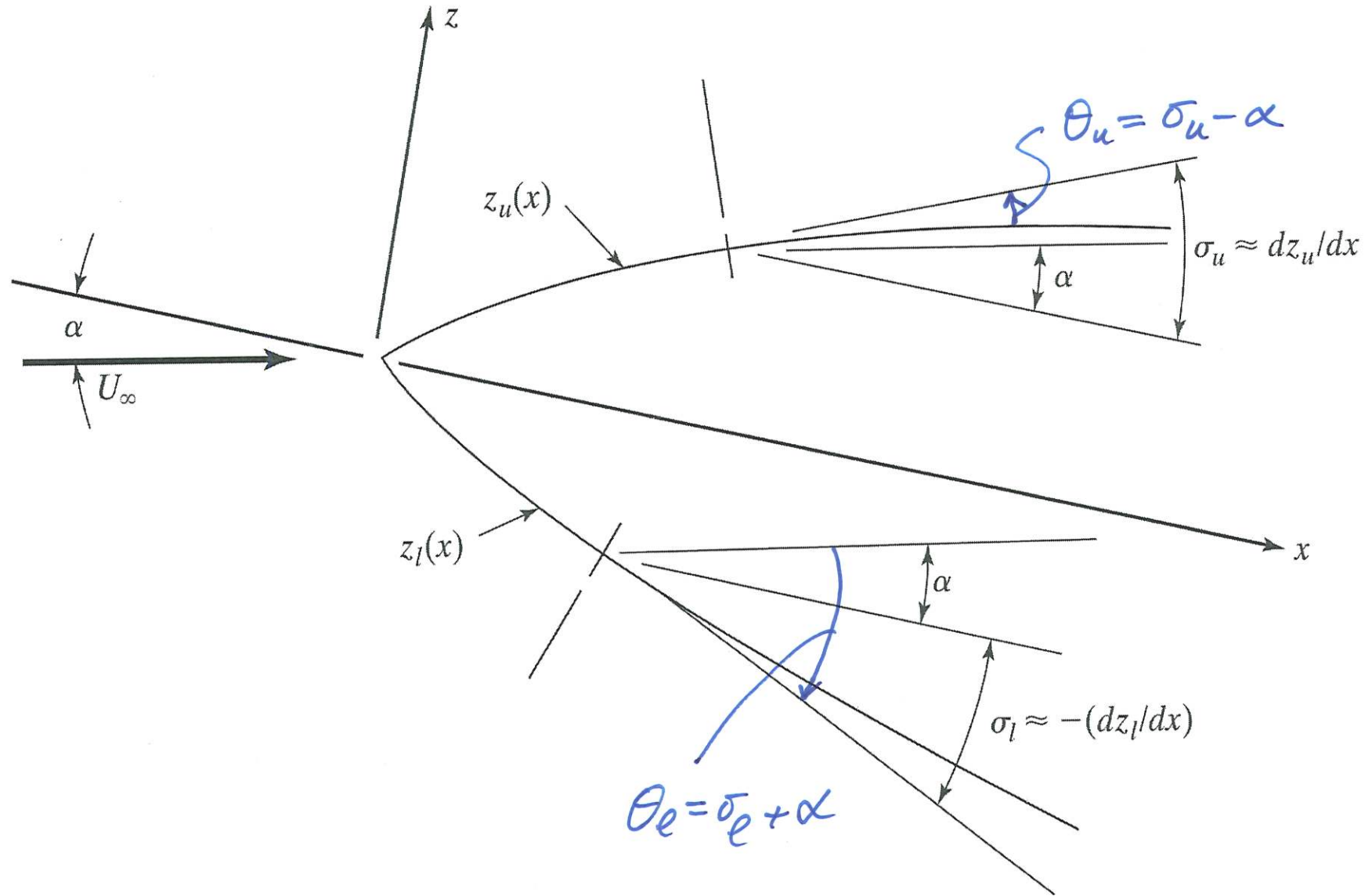
$$\Rightarrow C_p = \frac{2\theta}{\sqrt{M_\infty^2 - 1}} \quad \text{--- } \boxed{\text{rads}} \quad (10.1)$$

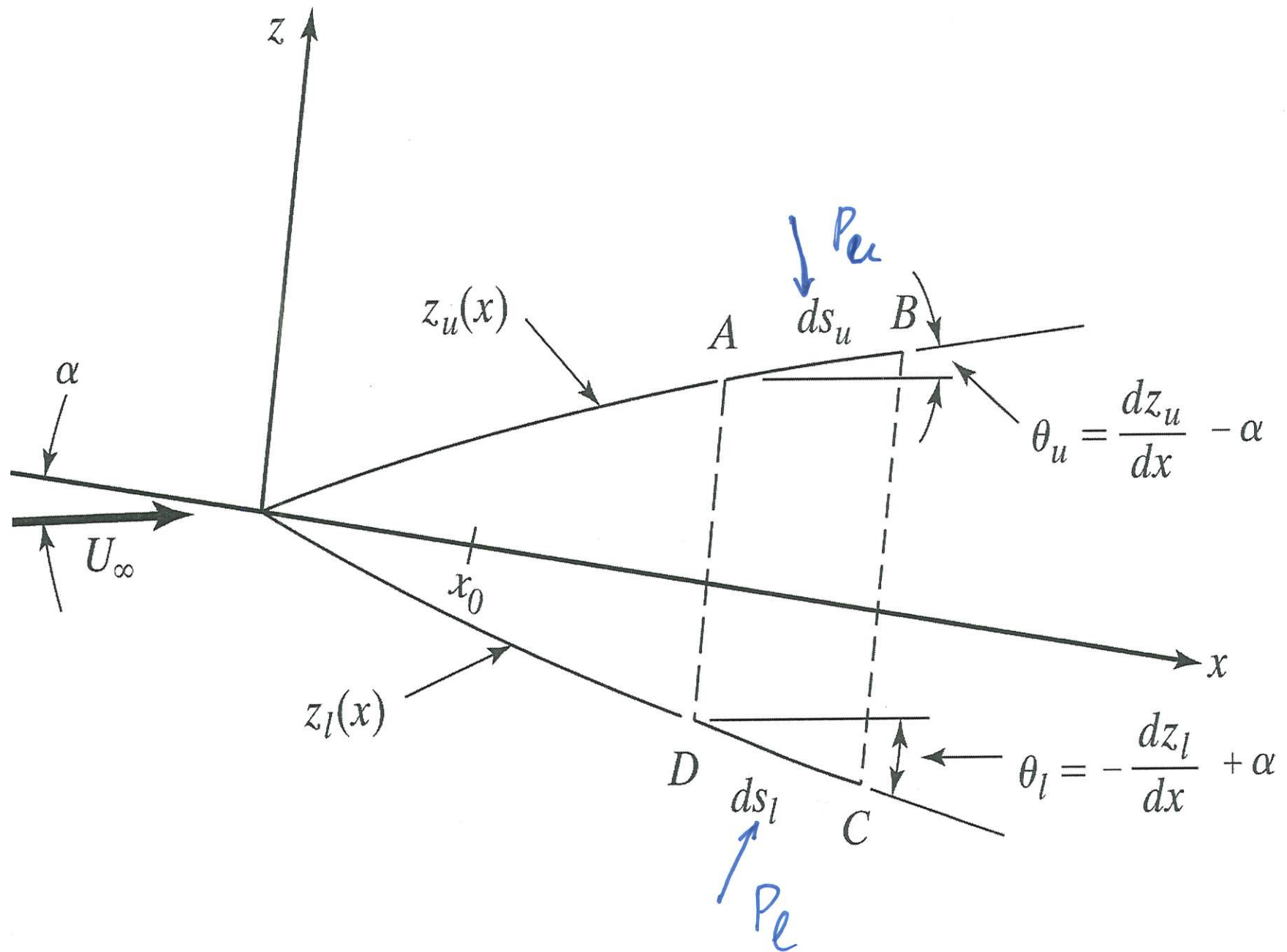
Sign of θ needs to be consistent; also

$$\theta_u = \frac{dz_u}{dx} - \alpha \quad \theta_l = -\frac{dz_l}{dx} + \alpha \quad (10.2)$$

upper
lower
wrt V_∞





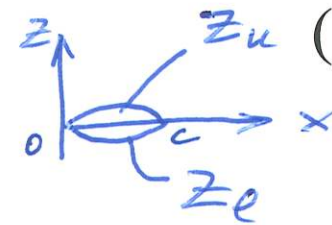


Lift

$$dl = p_l ds_l \cos \theta_l - p_u ds_u \cos \theta_u \Rightarrow dl = (p_l - p_u) dx \quad (10.4)$$

Nondimensionalizing: $dC_l = (C_{pl} - C_{pu}) d\left(\frac{x}{c}\right)$ (10.5)

$$dC_l = \frac{2}{\sqrt{M_\infty^2 - 1}} \left(2\alpha - \frac{dz_l}{dx} - \frac{dz_u}{dx} \right) d\left(\frac{x}{c}\right) \quad (10.6)$$



Integrating:

$$C_l = \frac{4\alpha \text{ rads}}{\sqrt{M_\infty^2 - 1}} \quad (10.8)$$

Lift curve slope: $\frac{dC_l}{d\alpha} = \frac{4}{\sqrt{M_\infty^2 - 1}} \quad (10.9)$

note: $\int_0^c \left(\frac{dz}{dx} \right)_{u,e} dx = 0$

Drag

(dD is a better notation, but we follow the text)

$$dd = p_l ds_l \sin \theta_l - p_u ds_u \sin \theta_u \quad (10.10)$$

$\sin \theta \approx \theta$

Small disturbances: $dd = p_l ds_l \theta_l - p_u ds_u \theta_u$ *($ds_l = ds_u \approx dx$)* (10.11)

Nondimensionalizing: *note: $\frac{p}{p_\infty} \theta / \frac{1}{2} \gamma M_\infty^2 = (\frac{p}{p_\infty} - 1) \theta / \frac{1}{2} \gamma M_\infty^2 + \theta / \frac{1}{2} \gamma M_\infty^2$*

$$dC_d = (C_{pl} \theta_l - C_{pu} \theta_u) d\left(\frac{x}{c}\right) + \frac{2}{\gamma M_\infty^2} (\theta_l - \theta_u) d\left(\frac{x}{c}\right) \quad (10.12)$$

$$dC_d = \frac{2}{\sqrt{M_\infty^2 - 1}} \left[2\alpha^2 + \left(\frac{dz_u}{dx}\right)^2 + \left(\frac{dz_l}{dx}\right)^2 \right] d\left(\frac{x}{c}\right) +$$

$$\left[\frac{-4\alpha}{\sqrt{M_\infty^2 - 1}} \left(\frac{dz_u}{dx} + \frac{dz_l}{dx}\right) + \frac{2}{\gamma M_\infty^2} \left(\frac{dz_u}{dx} - \frac{dz_l}{dx}\right) \right] d\left(\frac{x}{c}\right) \quad (10.13)$$

Integrating:

$$C_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}} + \frac{2}{\sqrt{M_\infty^2 - 1}} \int_0^1 \left[\left(\frac{dz_u}{dx} \right)^2 + \left(\frac{dz_l}{dx} \right)^2 \right] d\left(\frac{x}{c}\right) \quad (10.14)$$

For small angles

$$\frac{dz_u}{dx} = \tan \sigma_u \approx \sigma_u$$

Define: $\frac{1}{c} \int_0^c \left[\left(\frac{dz_u}{dx} \right)^2 \right] dx \equiv \bar{\sigma}_u^2$; $\frac{1}{c} \int_0^c \left[\left(\frac{dz_l}{dx} \right)^2 \right] dx \equiv \bar{\sigma}_l^2$ (10.15)

$d = d'$ (drag force per unit span)

$$C_d = \frac{d}{q_\infty c} = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[\alpha^2 + \frac{1}{2} (\bar{\sigma}_u^2 + \bar{\sigma}_l^2) \right] \quad (10.16)$$

An alternative formula using camber and thickness:

$$C_d = \frac{4}{\sqrt{M_\infty^2 - 1}} (\alpha^2 + \bar{\sigma}_c^2 + \bar{\sigma}_t^2)$$

Where: $z_c = (z_u + z_l) / 2$; $z_t = (z_u - z_l) / 2$
camber *thickness*

$$\frac{1}{c} \int_0^c \left[\left(\frac{dz_c}{dx} \right)^2 \right] dx \equiv \bar{\sigma}_c^2 ; \quad \frac{1}{c} \int_0^c \left[\left(\frac{dz_t}{dx} \right)^2 \right] dx \equiv \bar{\sigma}_t^2$$

In general: $C_d = C_{d, \text{due to lift}} + C_{d, \text{due to shape}} + C_{d, \text{viscous}}$ (10.17)
+ induced drag (3D)

Pitch Moment

$$dm_{x0} = (p_u - p_l)(x - x_o) dx \quad (10.18)$$

Nondimensionalizing:

$$dC_{m_{xo}} = (C_{pu} - C_{pl}) \left(\frac{x - x_o}{c} \right) d \left(\frac{x}{c} \right) \quad (10.19)$$

using (10.1)

$$dC_{m_{xo}} = \frac{2}{\sqrt{M_\infty^2 - 1}} (\theta_u - \theta_l) \left(\frac{x - x_o}{c} \right) d \left(\frac{x}{c} \right) \quad (10.20)$$

$C_p = \frac{2\theta}{\sqrt{M_\infty^2 - 1}}$

$$C_{m_{xo}} = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[\alpha \left(\frac{x_o}{c} - \frac{1}{2} \right) + \bar{m}_{c_{xo}} \right] \quad (10.22)$$

$$\bar{m}_{c_{xo}} = \int_0^1 \frac{dz_c}{dx} \left(\frac{x - x_o}{c} \right) d \left(\frac{x}{c} \right)$$

For leading edge moment:

$$C_{m\ le} = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} \left(-\frac{1}{2} + \bar{m}_{cle} \right)$$

Center of pressure:

effect of camber

$$\frac{x_{cp}}{c} = -\frac{C_{m\ l.e.}}{C_n} = \frac{(2\alpha - 4\bar{m}_{cle}) / \sqrt{M_\infty^2 - 1}}{4\alpha / \sqrt{M_\infty^2 - 1}} = \frac{1}{2} - \frac{\bar{m}_{cle}}{\alpha}$$

Aerodynamic Center:

need to cancel to be independent of α

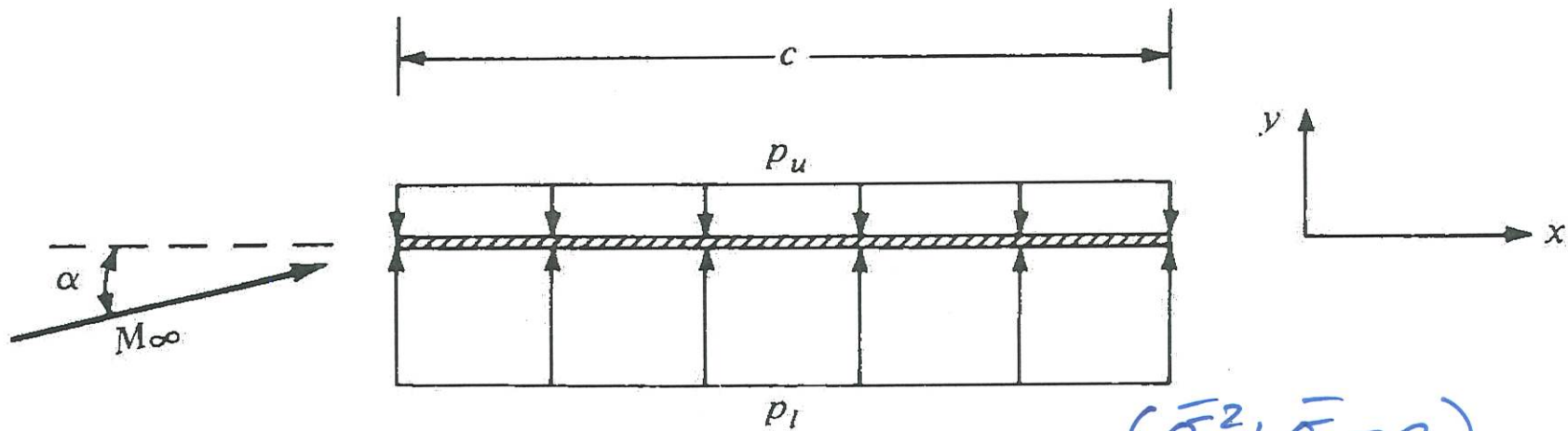
$$C_{m\ a.c.} = C_{m\ l.e.} + \frac{x_{ac}}{c} C_l = \frac{1}{\sqrt{M_\infty^2 - 1}} \left(-2\alpha + 4\bar{m}_{cle} + \frac{x_{ac}}{c} 4\alpha \right)$$

Therefore: $x_{ac} = c/2$ and $C_{m\ a.c.} = \frac{4\bar{m}_{cle}}{\sqrt{M_\infty^2 - 1}}$

(for incompressible: $x_{ac} = \frac{c}{4}$)

Examples

Flat plate at a small angle of attack α



$$C_\ell = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$$

$$C_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}}$$

$$C_{m \text{ l.e.}} = -\frac{2\alpha}{\sqrt{M_\infty^2 - 1}} \quad (\bar{\sigma}_c^2 + \bar{\sigma}_t = 0)$$

Center of pressure: $\frac{x_{cp}}{c} = \frac{1}{2} \quad (\text{no camber})$

Example

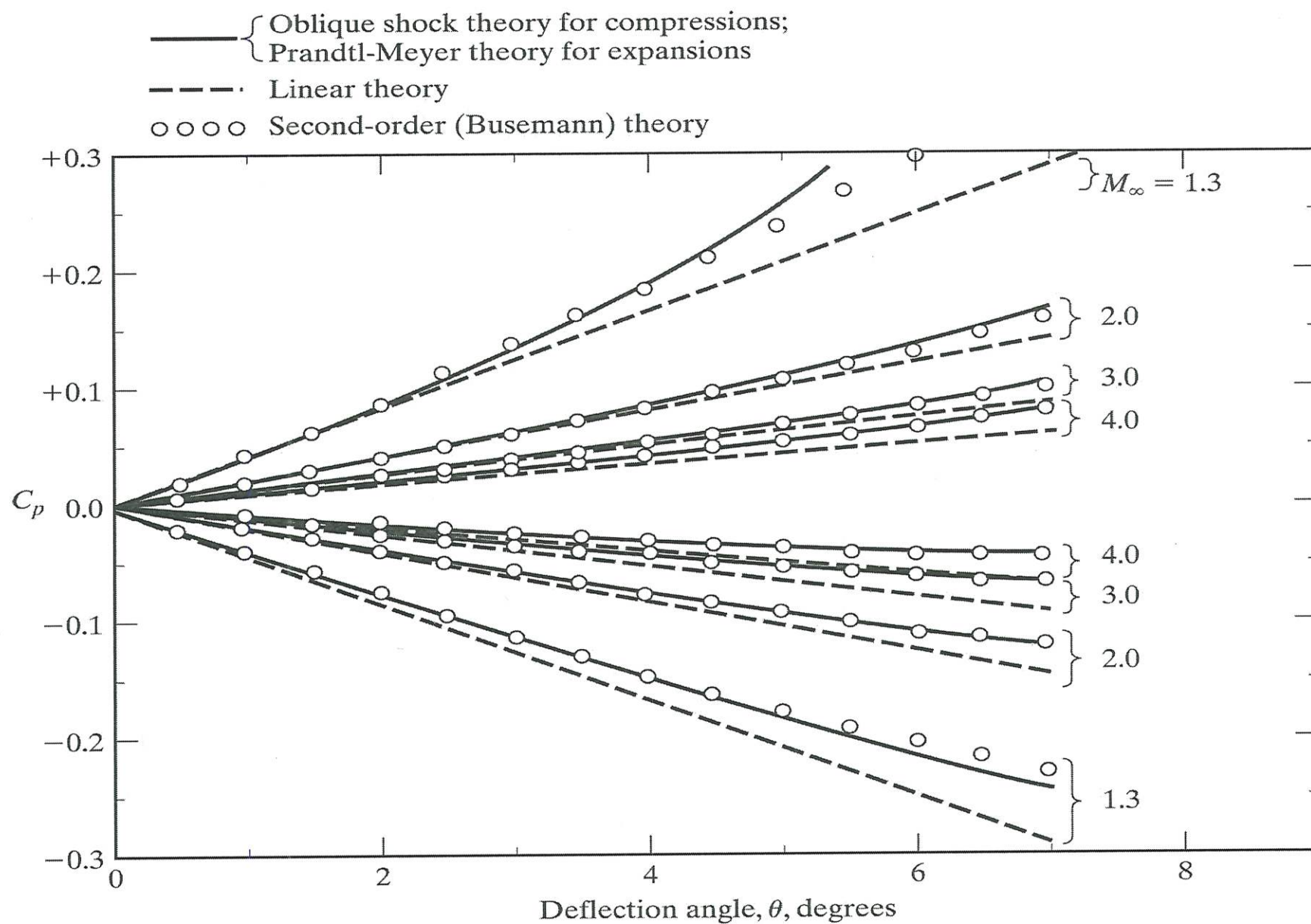
Using linearized theory find C_ℓ , C_d for a flat plate at $M = 2.6$ at $\alpha = 5^\circ, 15^\circ, 30^\circ$ and compare with shock expansion theory

	<u>linear theory</u>	<u>exact theory</u> (oblique shock, expansion fan)	<u>error</u>
a) $\alpha = 5^\circ$			
$C_\ell = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}} = 0.1455$		0.148	1.69%
$C_d = \frac{4\alpha^2}{\sqrt{M_\infty^2 - 1}} = C_\ell \alpha = 0.0127$		0.0129	1.53%

b) $\alpha = 15^\circ$	$C_\ell =$	0.436	0.452	3.41%
	$C_d =$	<u>0.114</u>	0.121	5.7%
c) $\alpha = 30^\circ$	$C_\ell =$	0.873	1.19	26.7%
	$C_d =$	<u>0.457</u>	0.687	33.5%

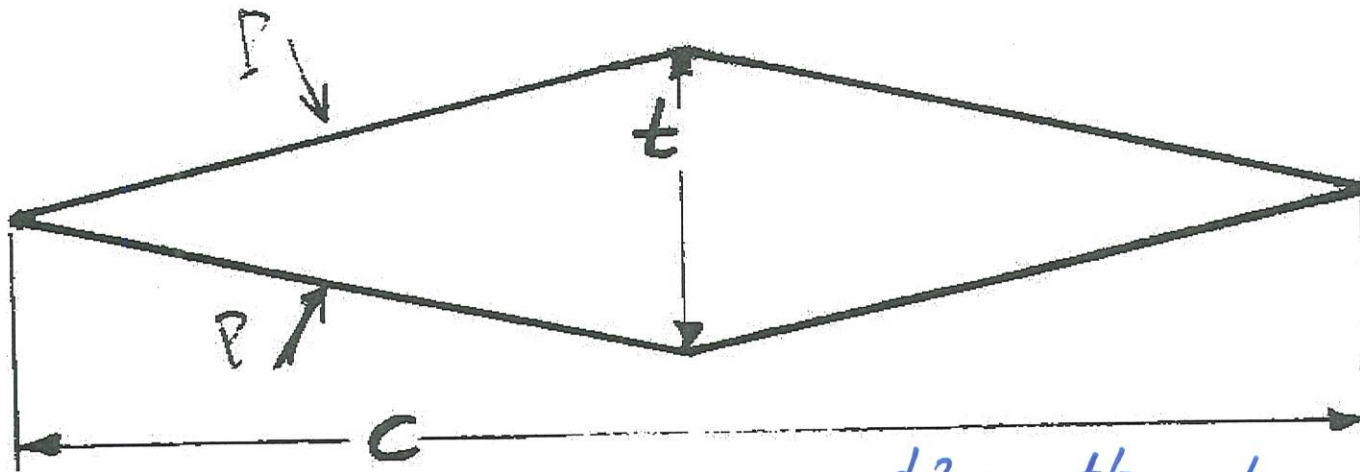
Conclusion: at low α linear theory is good; its accuracy deteriorates with high α , as expected.

[Repeat this example for $M=5$ (validity limit of linearized theory) and check results]



Compare the C_d of a diamond shape and a parabolic (biconvex)

airfoil $\left(Y = \pm 2t \left(\frac{x}{c} - \frac{x^2}{c^2} \right) \right)$ for the same max thickness t at $\alpha = 0$;

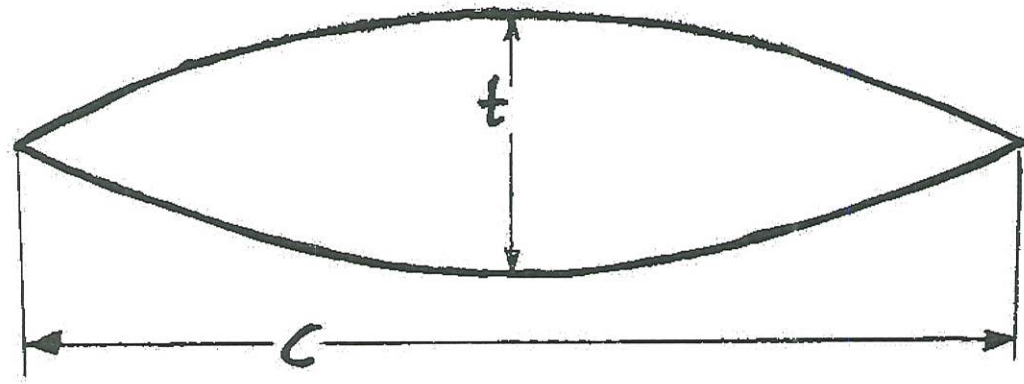


$$\left| \frac{dz}{dx} \right| = \frac{t/2}{c/2} = \frac{t}{c}$$

$$C_d = \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\underset{=0}{\bar{\sigma}_c^2} + \bar{\sigma}_t^2 \right) = \frac{1}{c} \frac{4}{\sqrt{M_\infty^2 - 1}} \int_0^c \underbrace{\left(\frac{dz}{dx} \right)^2}_{\text{thickness}} dx = \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\frac{t}{c} \right)^2$$

$$M_\infty \geq 1$$

$$\alpha = 0$$



$$z(x) = \pm 2t \left(\frac{x}{c} - \frac{x^2}{c^2} \right) \Rightarrow \frac{dz}{dx} = \pm \frac{2t}{c} \left(1 - \frac{2x}{c} \right)$$

upper (for +) and *lower* (for -)

$$C_d = \frac{1}{c} \frac{4}{\sqrt{M_\infty^2 - 1}} \int_0^c \left(\frac{dz}{dx} \right)^2 dx = \frac{1}{c} \frac{4}{\sqrt{M_\infty^2 - 1}} \int_0^c \frac{4t^2}{c^2} \left(1 - \frac{4x}{c} + \frac{4x^2}{c^2} \right) dx$$

thickness

$$= \frac{1}{c} \frac{4}{\sqrt{M_\infty^2 - 1}} \cdot \frac{4t^2}{c^2} \left[c - \frac{4c^2}{2c} + \frac{4}{c^2} \frac{c^3}{3} \right]$$

(camber = 0)

$$= 1 - 2 + \frac{4}{3} = \frac{1}{3}$$

$\Rightarrow$ parabolic airfoil has 33% more drag at $\alpha = 0$

$$= \frac{4}{\sqrt{M_\infty^2 - 1}} \left(\frac{4}{3} \right) \left(\frac{t}{c} \right)^2$$

at $\alpha \neq 0$?

$$C_\ell = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$$

$$C_{d \downarrow \text{diamond}} = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[\alpha^2 + \left(\frac{t}{c} \right)^2 \right]$$

effect of α

effect of thickness

$$C_{d \downarrow \text{parabolic}} = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[\alpha^2 + \frac{4}{3} \left(\frac{t}{c} \right)^2 \right]$$

L/D is independent of M_∞ :

(camber is 0 in both cases)

parabolic:

$$\frac{C_\ell}{C_d} = \frac{\alpha}{\alpha^2 + \frac{4}{3} \left(\frac{t}{c} \right)^2}$$

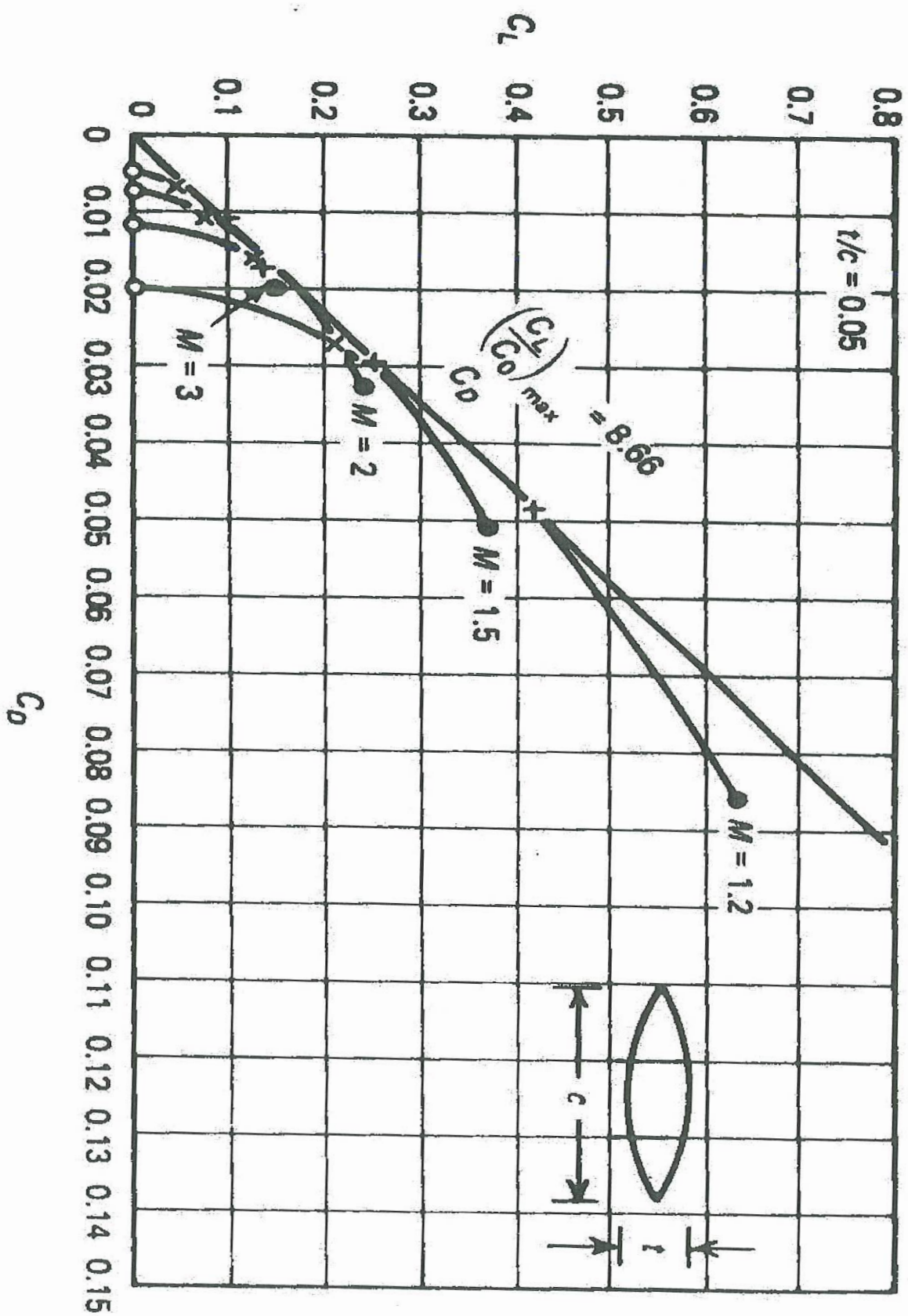
$$\max \frac{C_\ell}{C_d} \text{ when } \alpha = \sqrt{\frac{4}{3}} \frac{t}{c} \Rightarrow \left(\frac{C_\ell}{C_d} \right)_{\max} = \frac{1}{\sqrt{\frac{16}{3}} \left(\frac{t}{c} \right)}$$

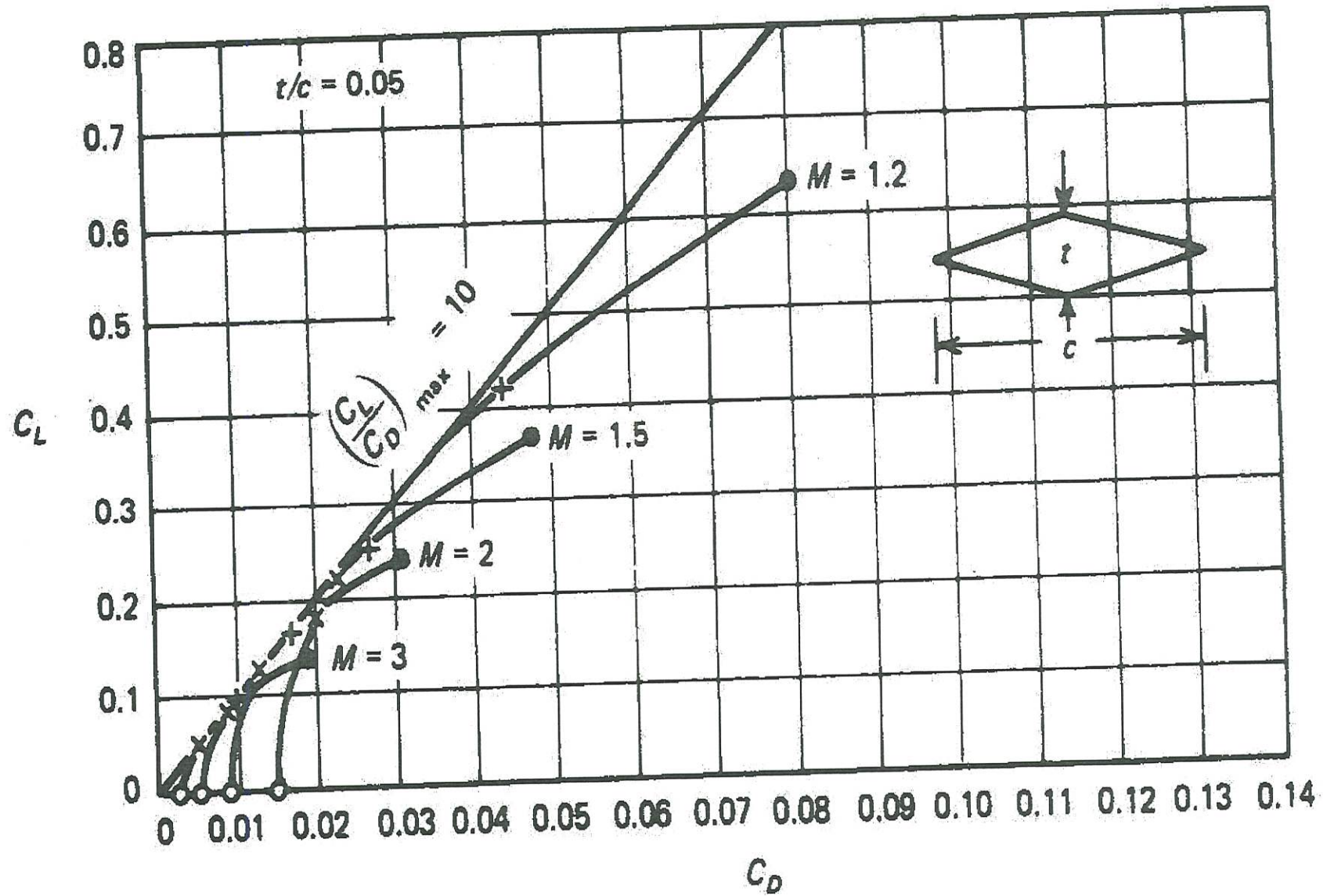
(take $\frac{\partial (C_\ell/C_d)}{\partial \alpha} = 0 \Rightarrow \dots$)

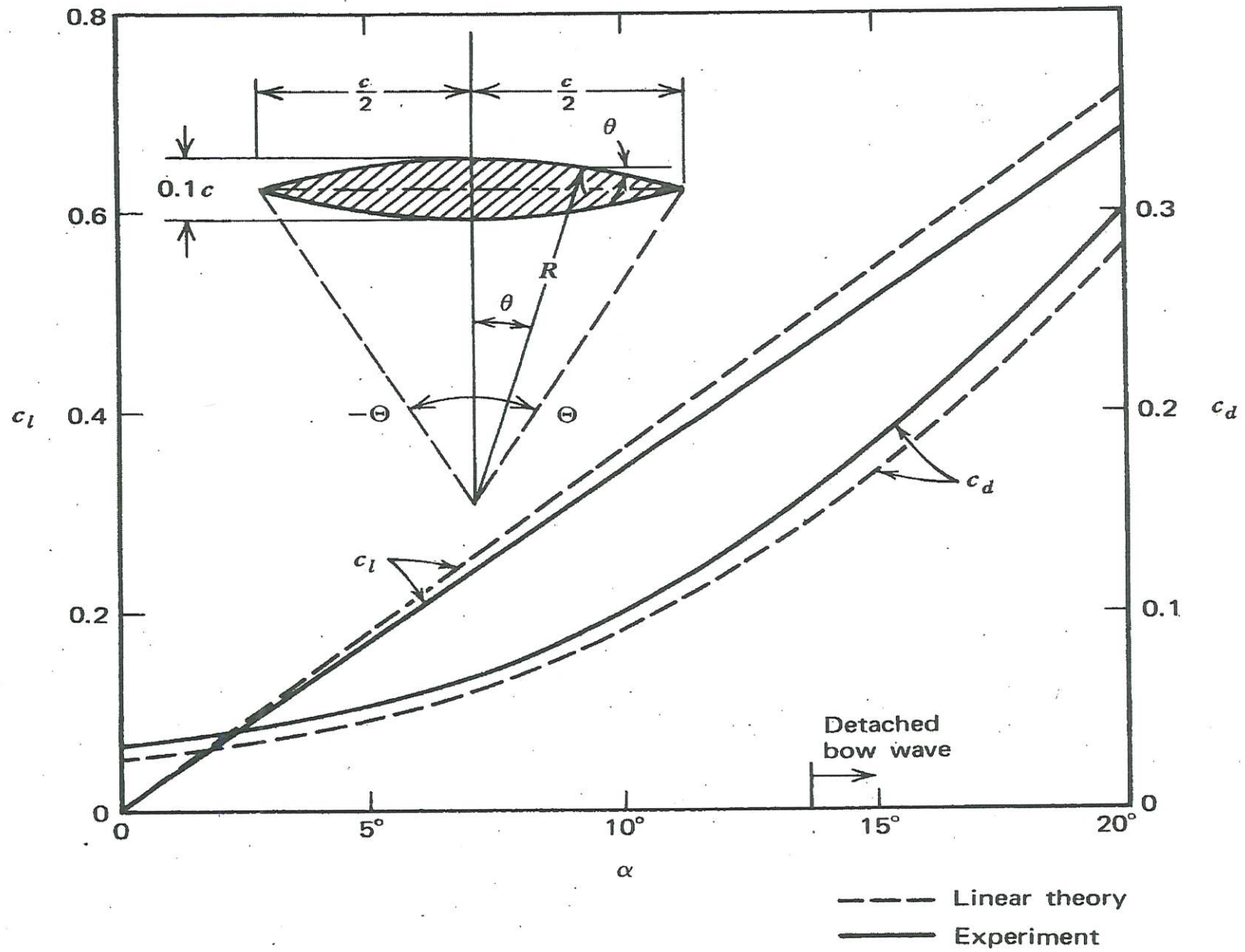
diamond:

$$\frac{C_\ell}{C_d} = \frac{\alpha}{\alpha^2 + \left(\frac{t}{c} \right)^2}$$

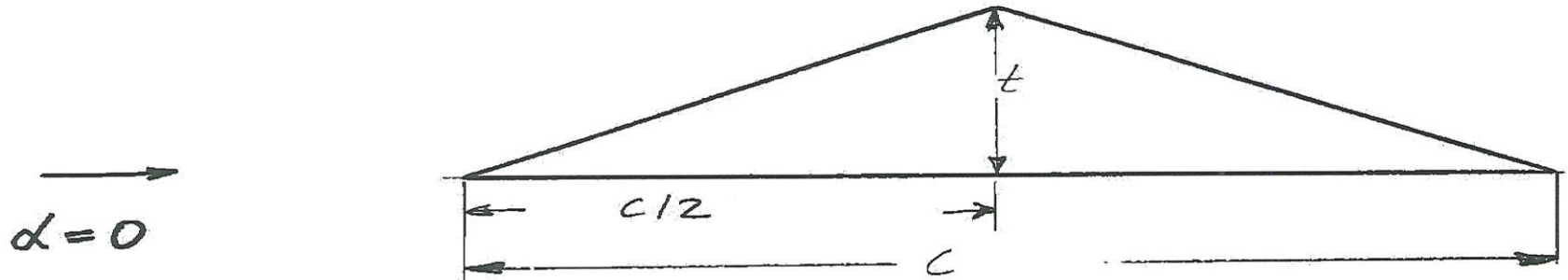
$$\max \frac{C_\ell}{C_d} \text{ when } \alpha = \frac{t}{c} \Rightarrow \left(\frac{C_\ell}{C_d} \right)_{\max} = \frac{1}{2 \left(\frac{t}{c} \right)}$$







Example: triangular airfoil



$$C_d = \frac{2}{\sqrt{M_\infty^2 - 1}} (\bar{\sigma}_u^2 + \bar{\sigma}_t^2) ; \quad \bar{\sigma}_u^2 = \frac{1}{c} \int_0^c \left(\frac{dz_u}{dx} \right)^2 dx = \frac{1}{c} \left[\int_0^{c/2} \left(\frac{t}{c/2} \right)^2 + \int_{c/2}^c \left(\frac{-t}{c/2} \right)^2 \right] dx$$

$$\bar{\sigma}_e^2 = 0 \quad = \frac{2}{c} \left[\int_0^{c/2} \left(\frac{t}{c/2} \right)^2 dx \right] = 4 \left(\frac{t}{c} \right)^2$$

$$\Rightarrow C_d = \frac{4}{\sqrt{M_\infty^2 - 1}} \left[2 \left(\frac{t}{c} \right)^2 \right]$$

$\Rightarrow C_d$ of triangle is higher than the C_d of diamond

note: $c_e = 0$ for $\alpha = 0$ ($c_e = \frac{4\alpha}{\sqrt{M_\infty^2 - 1}}$) although the airfoil is cambered